

# Resultater fra QH3 for Hilde Celine Løvland

Resultat for dette forsøket: 21 av 21

Innlevert 17. sep. i 17.47

Dette forsøket tok 26 minutter

Riktig svar



Spørsmål 1

1 / 1 poeng

Hydraulic orifice . Refer to Figure 3.1 in support material for QH3 . Liquid density ( $\rho$ ) = 883 kg/m<sup>3</sup> . Discharge coefficient ( $C_d$ ) = 0.84 . Discharge area ( $A_d$ ) = 11 mm<sup>2</sup> . Pressure ( $p_B$ ) = 9 bar . Pressure ( $p_A$ ) = 61 bar . What is the flow ( $Q$ ) in l/min?

60,1672

60,1672 (med marginen: 0,6016)

Riktig svar



Spørsmål 2

1 / 1 poeng

Directional control valve . Refer to Figure 3.2 in support material for QH3 . A 4/3-way directional control valve is in position "a" . The 4/3-way directional control valve has identical discharge coefficients:  $C_{dPA} = C_{dBT} = C_{dPB} = C_{dAT} = C_d$  . The 4/3-way directional control valve has identical discharge areas:  $A_{dPA} = A_{dBT} = A_{dPB} = A_{dAT} = A_d$  . Liquid density ( $\rho$ ) = 872 kg/m<sup>3</sup> . Discharge coefficient ( $C_d$ ) = 0.72 . Discharge area ( $A_d$ ) = 16 mm<sup>2</sup> . Flow ( $Q_{PA}$ ) = 20 l/min . What is the pressure difference ( $dp_{PA}$ ) in bar?

3,6504

3,6504 (med marginen: 0,0365)

Riktig svar



Spørsmål 3

1 / 1 poeng

Proportional directional control valve . Refer to Figure 3.3 in support material for QH3 . The 4/3-way proportional directional control valve is symmetric:  $C_{dPA} = C_{dBT} = C_{dPB} = C_{dAT} = C_d$  . The 4/3-way proportional directional control valve is symmetric:  $A_{dMax,PA} = A_{dMax,BT} = A_{dMax,PB} = A_{dMax,AT} = A_{dMax}$  . Liquid density ( $\rho$ ) = 873 kg/m<sup>3</sup> . Discharge coefficient ( $C_d$ ) = 0.88 . Max valve spool travel ( $x_{Max}$ ) = 10.0 mm . Maximum discharge area ( $A_{dMax}$ ) = 12 mm<sup>2</sup> . Flow ( $Q_{PA}$ ) = 28 l/min . Pressure ( $p_P$ ) = 17 bar . Pressure ( $p_A$ ) = 6 bar . Pressure ( $p_B$ ) = 8 bar . Pressure ( $p_T$ ) = 0 bar . What is the spool travel ( $x$ ) in mm? (remember the sign +/-)

8,8032 (med marginen: 0,088)

Riktig svar



Spørsmål 4

5 / 5 poeng

Basic valve characteristics . Refer to Figure 3.4d in support material for QH3 . Consider a pressure reduction valve in series with an orifice . Liquid density ( $\rho$ ) = 873 kg/m<sup>3</sup> . Discharge coefficient ( $C_{d1}$ ) = 0.94 . Discharge area ( $A_{d1}$ ) = 19 mm<sup>2</sup> . Discharge coefficient ( $C_{d2}$ ) = 0.94 . Discharge area ( $A_{d2}$ ) = 33 mm<sup>2</sup> . Rated pressure ( $p_0$ ) = 10 bar . Rated flow ( $Q_0$ ) = 40 l/min . Close pressure ( $p_{cl}$ ) = 50 bar . Flow ( $Q_{in}$ ) = 119 l/min . Pressure ( $p_T$ ) = 0 bar . What is the pressure ( $p_B$ ) in bar?

53,8286 (med marginen: 0,5382)

Riktig svar



Spørsmål 5

3 / 3 poeng

4/3 directional control valve and ideal hydraulic cylinder . Refer to Figure 3.5 (left) in support material for QH3 . The valve A-port is connected to the cylinder piston side . The valve B-port is connected to the cylinder rod side . The 4/3-way directional control valve is in position "a" . The 4/3-way directional control valve has identical discharge coefficients:  $C_{dPA} = C_{dBT} = C_{dPB} = C_{dAT} = C_d$  . The 4/3-way directional control valve has symmetrical discharge areas:  $A_{dPA} = A_{dPB}$  and  $A_{dBT} = A_{dAT}$  . The cylinder is extracting . The cylinder is subjected to a resistive (positive) load . Cylinder diameter ( $d$ ) = 50 mm . Cylinder rod diameter ( $d_R$ ) = 25 mm . Cylinder load force ( $F_L$ ) = 18.0 kN . Liquid density ( $\rho$ ) = 878 kg/m<sup>3</sup> . Discharge coefficient ( $C_d$ ) = 0.72 . Flow ( $Q_{in}$ ) = 21 l/min . Discharge area ( $A_{dPA}$ ) = 9.9 mm<sup>2</sup> . Discharge area ( $A_{dBT}$ ) = 6.6 mm<sup>2</sup> . What is the cylinder velocity ( $v$ ) in mm/s?

178,2535 (med marginen: 1,7826)

Riktig svar



Spørsmål 6

3 / 3 poeng

4/3 directional control valve and ideal hydraulic motor . Refer to Figure 3.6 (right) in support material for QH3 . The 4/3-way directional control valve is in position "b" . The 4/3-way directional control valve has identical discharge coefficients:  $C_{dPA} = C_{dBT} = C_{dPB} = C_{dAT} = C_d$  . The 4/3-way directional control valve has symmetrical

discharge areas:  $A_{dPA} = A_{dPB}$  and  $A_{dBT} = A_{dAT}$ . The motor is subjected to an assistive (negative) load. Motor displacement ( $D$ ) = 250 cm<sup>3</sup>/rev. Motor torque load ( $M_L$ ) = 556. Liquid density ( $\rho$ ) = 875 kg/m<sup>3</sup>. Discharge coefficient ( $C_d$ ) = 0.74. Flow ( $Q_{in}$ ) = 260 l/min. Discharge area ( $A_{dPB}$ ) = 78 mm<sup>2</sup>. Discharge area ( $A_{dAT}$ ) = 31 mm<sup>2</sup>. What is the pressure ( $p_P$ ) in bar?

41,0323

41,0323 (med marginen: 0,4103)

Riktig svar



Spørsmål 7

3 / 3 poeng

Hydraulic base unit. Refer to Figure 3.7b in support material for QH3. Consider a 4/3-way proportional DCV and an extracting cylinder. The 4/3-way proportional directional control valve is symmetric:  $C_{dPA} = C_{dBT} = C_{dPB} = C_{dAT} = C_d$ . The 4/3-way proportional directional control valve has symmetrical discharge areas:  $A_{dMax,PA} = A_{dMax,PB}$  and  $A_{dMax,BT} = A_{dMax,AT}$ . Liquid density ( $\rho$ ) = 882 kg/m<sup>3</sup>. Discharge coefficient ( $C_d$ ) = 0.78. Maximum discharge area ( $A_{dMax,PA}$ ) = 26 mm<sup>2</sup>. Maximum discharge area ( $A_{dMax,BT}$ ) = 29 mm<sup>2</sup>. Max valve spool travel ( $x_{Max}$ ) = 10.0 mm. Valve spool travel ( $x$ ) = 6.7 mm. The cylinder is extracting. The cylinder is subjected to a resistive (positive) load. Cylinder diameter ( $d$ ) = 80 mm. Cylinder rod diameter ( $d_R$ ) = 50 mm. Cylinder hydromechanical efficiency ( $\eta_{hmC}$ ) = 0.96. Cylinder load force ( $F_L$ ) = 59.8 kN. Flow ( $Q_{in}$ ) = 33 l/min. What is the pressure ( $p_B$ ) in bar?

2,1567

2,1567 (med marginen: 0,0216)

Riktig svar



Spørsmål 8

4 / 4 poeng

Hydraulic base unit. Refer to Figure 3.8a in support material for QH3. Consider a 4/3-way proportional DCV, a counterbalance valve, and a motor. The 4/3-way proportional directional control valve is symmetric:  $C_{dPA} = C_{dBT} = C_{dPB} = C_{dAT} = C_d$ . The 4/3-way proportional directional control valve has symmetrical discharge areas:  $A_{dMax,PA} = A_{dMax,PB}$  and  $A_{dMax,BT} = A_{dMax,AT}$ . Liquid density ( $\rho$ ) = 873 kg/m<sup>3</sup>. Discharge coefficient ( $C_d$ ) = 0.93. Maximum discharge area ( $A_{dMax,PA}$ ) = 39 mm<sup>2</sup>. Maximum discharge area ( $A_{dMax,BT}$ ) = 38 mm<sup>2</sup>. Max valve spool travel ( $x_{Max}$ ) = 10.0 mm. Valve spool travel ( $x$ ) = 4.9 mm. The motor is subjected to an assistive (negative) load. Motor displacement ( $D$ ) = 125 cm<sup>3</sup>/rev. Motor volumetric efficiency ( $\eta_{vM}$ ) = 0.96. Motor hydromechanical efficiency ( $\eta_{hmM}$ ) = 0.98. Motor load torque ( $M_L$ ) = 275 N\*m. Pilot area ratio ( $\alpha$ ) = 2.4. Crack pressure ( $p_{cr}$ ) = 230 bar. Flow ( $Q_{in}$ ) = 45 l/min. What is the motor speed ( $n$ ) in rev/min?

375

375 (med marginen: 3,75)

Quizresultat: 21 av 21